

$$p^x (1-x)^{1-y}$$

We know  $r=2$  and  $k=2$

$$p^{\frac{1}{2}} (1-\frac{1}{2})^{\frac{1}{2}}$$

$$p^{\frac{1}{2}} (\frac{1}{2})^{\frac{1}{2}} = 0.0388$$

$$p = 0.197$$

$$0.197^{\frac{1}{2}} = 0.651$$

$$0.651^{\frac{1}{2}} = 0.038$$